

Supply Chain Analytics: Phase 3 – Risk Modeling & Financial Exposure

Project Overview

Phase 3 is the "Intelligence Layer" of the project. Here, we merge real-time inventory levels with historical supplier performance to create a predictive risk model. This phase moves beyond looking at what happened (descriptive) to identifying where the supply chain is most likely to fail next (predictive/prescriptive).

1. Predictive Risk Framework

The core innovation in this phase is the transition from static database flags to **Dynamic Risk Scoring**.

Live Inventory Sanity Check

We implemented vw_inventory_risk_base to bypass potentially stale system flags. By calculating is_below_reorder_level mathematically (\$Quantity < Reorder Level\$), we ensure that the "Red Zone" reports are based on actual stock levels rather than legacy data status.

The "Alphabet Fix" & Risk Aggregation

We developed a robust aggregation method in vw_product_supplier_risk_agg to solve the "Alphabet Trap" (where 'Medium' risk might hide 'High' risk in alphabetical sorting).

- **The Logic:** We converted categorical risk tiers into numerical weights (\$High=3, Medium=2, Low=1\$) before taking the MAX. This ensures that a product supplied by multiple vendors is always flagged by its **weakest link**.

2. The Composite Risk Score (CRS)

We engineered a proprietary **Composite Risk Score** ranging from **0 to 5** to prioritize management intervention:

$$Score = (\text{Stockout Factor} \times 2) + (\text{Supplier Risk Score})$$

- **Stockout Factor (0 or 2):** A binary multiplier. If the item is under the reorder level, it immediately gains 2 points.
- **Supplier Risk (1–3):** Adds the severity of the delay risk from the associated suppliers.

Score	Status	Action Required

5	CRITICAL	Stockout + High-Risk Supplier. Immediate replenishment/alternative sourcing needed.
3-4	ELEVATED	Partial Stockout or Low Stock + Unreliable Supplier. Monitor closely.
1-2	STABLE	Healthy stock levels or reliable supply chain partners.

3. Financial Impact & Business Value

To bridge the gap between logistics and finance, we developed the vw_financial_risk_exposure view.

- **Revenue at Risk:** Instead of just reporting "low stock," we quantify the crisis in dollars: $\text{\$} \times \text{\$}$
- **Warehouse Exposure:** Aggregates risk at a site level, allowing the business to see which physical locations are becoming "high-risk zones" due to poor supplier fulfillment.

4. Key Insights Generated

- **The Red Zone Count:** [e.g., 3 products currently at Score 5]
- **Total Revenue at Risk:** [22624.83 \$]
- **Most Vulnerable Warehouse:** [3] proved to be the most exposed warehouse out of the three main warehouses

5. Strategic Recommendations

1. **Diversification:** For items with a CRS of 5, initiate immediate "dual-sourcing" to find a backup supplier with a 'Low' risk tier.
2. **Buffer Adjustment:** Increase safety stock levels for products where the max_delay_risk_tier is consistently 'High'.
3. **Financial Prioritization:** Use the revenue_at_risk metric to decide which purchase orders to expedite via air freight vs. standard shipping.